

Graph-Aware Deep Reinforcement Learning for Adaptive Capacity Planning in Distributed Personalized Regenerative Medicine Manufacturing Networks

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Abstract

Autologous cell therapy and other personalized regenerative medicine (PRM) products are manufactured on demand across distributed clinical networks, where each product is *perishable*, *identity-bound* to a single patient and non-fungible, and needed only while that patient remains clinically eligible. These characteristics couple capacity, inventory, and patient outcomes into one networked decision. We formulate distributed PRM capacity planning as a constrained graph Markov decision process and release a reusable simulation testbed with patient-condition dynamics, expiry-aware inventory, geographic transfer delays, and resource sharing restricted to fungible reagents and idle bioreactor capacity. To solve it, we develop an *advantage-filtered residual graph convolutional network-deep deterministic policy gradient* controller (AFR-GCN-DDPG). The study-specific AFR label denotes the complete anchor-plus-residual controller, in which a graph convolutional state encoder learns bounded corrections to a transparent two-period look-ahead

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policy; it is not a new actor–critic backbone. In a five-seed demand-prior-drift experiment, with 300 training episodes and 100 paired holdout replications per seed, AFR-GCN-DDPG reduced the total weighted objective by 0.016360% relative to MDL-2, with a two-level 95% confidence interval fully below zero. A matched flat residual DDPG policy also improved on MDL-2, while the graph-versus-flat interval crossed zero, so this replenishment-only experiment supports residual learning but not a graph-specific gain. After extending the residual action to geographic reagent and idle-capacity transfers, a three-seed abrupt-regional-shift experiment reduced the objective by 1.1337% relative to MDL-2 and by 1.1166% relative to a parameter-matched flat policy while improving completion service. A separate conservative online-TD3 screen produced a final graph policy that improved on MDL-2 and matched flat TD3; however, its pooled final-versus-frozen-pretraining interval crossed zero. The evidence therefore supports graph-aware anchored control under geographically structured non-stationarity, while the incremental contribution of online actor–critic updates remains an open attribution question.

Keywords: Graph convolutional networks, Deep reinforcement learning, Actor–critic methods, Capacity planning, Distributed manufacturing, Personalized regenerative medicine, Perishable inventory, Transshipment

1. Introduction

Distributed make-to-order manufacturing networks are increasingly important in engineering systems where products are individualized, production capacity is scarce, and service outcomes depend on time-sensitive sequential

5 decisions. Personalized regenerative medicine (PRM), and autologous cell
6 therapy in particular, is a demanding instance of this class. Each therapy is
7 manufactured from an individual patient’s own biological material, so pro-
8 duction can begin only when that patient’s specimen, an available bioreactor,
9 and sufficient consumables are simultaneously present, and the finished prod-
10 uct must be returned to the same patient within a limited viability window.
11 Because the product is patient-specific, it cannot be pre-produced, pooled, or
12 substituted, and the opportunity to buffer against uncertainty with inventory
13 is fundamentally limited [1, 2]. Three properties therefore distinguish PRM
14 capacity planning from conventional make-to-order or inventory control. The
15 product is *perishable*, subject to a hard shelf-life and vein-to-vein deadline; it
16 is *identity-bound*, tied one-to-one to a single patient and non-fungible across
17 patients; and its demand is driven by *patient health*, since a patient whose
18 condition deteriorates while waiting may become clinically ineligible before
19 treatment can be delivered. These are not marginal concerns. Autologous
20 CAR-T manufacturing has a turnaround measured in weeks, an estimated 4–
21 7% of patients never receive their product because of manufacturing failure,
22 and in multiple myeloma roughly one quarter of patients die while waitlisted
23 for a BCMA-directed CAR-T production slot [3]. Discrete-event simulation
24 links longer waits directly to survival, with predicted one-year mortality in
25 large B-cell lymphoma rising from about 36% to 76% as vein-to-vein time
26 increases from one to nine months [4]. With the CAR-T market projected to
27 reach tens of billions of US dollars by the early 2030s [5], delivering perish-
28 able, patient-specific product before patients deteriorate is a consequential
29 operational challenge.

30 These properties make capacity planning a coupled, network-level deci-
 31 sion problem rather than a collection of independent local ones. Locating
 32 production near patients through distributed or point-of-care manufacturing
 33 shortens fulfillment time, but it also creates spatial dependencies. A shortage
 34 at one clinic can sometimes be relieved by relocating idle bioreactor capacity
 35 or transshipping consumables from connected facilities, and each such action
 36 changes the future state of the whole network [6, 7]. The identity-bound
 37 nature of the product constrains what may be shared, because reagents and
 38 bioreactor capacity are fungible and can be rebalanced across the network,
 39 whereas a patient’s specimen and finished product are not. Coordination
 40 must therefore operate on shared resources while honoring a non-fungible,
 41 perishable, patient-indexed demand process. The state of such a system is
 42 naturally relational, since each facility is a node whose value depends on its
 43 neighbors’ queues and inventories and on the links along which shortages,
 44 slack, and disruptions propagate. A flat feature vector encodes local condi-
 45 tions but discards this topology, which motivates a graph-structured state
 46 representation as a matter of substance rather than convenience.

47 Reinforcement learning (RL) is a natural fit because the problem is se-
 48 quential. An action taken in one period alters future inventories, capacity,
 49 waiting specimens, and service outcomes. Continuous-control actor–critic
 50 methods, including deep deterministic policy gradient (DDPG) [8], twin de-
 51 layed DDPG (TD3) [9], soft actor–critic (SAC) [10], and proximal policy op-
 52 timization (PPO) [11], are standard tools for such real-valued control. Graph
 53 convolutional networks (GCNs) [12] provide a permutation-equivariant rela-
 54 tional inductive bias for encoding networked state. This machinery is by now

55 well established, and a substantial literature already couples graph learning
 56 with reinforcement learning for supply-chain and resource-allocation prob-
 57 lems [13, 14, 15, 16, 17]. We therefore do not claim novelty in combining
 58 graph representation with reinforcement learning.

59 What that literature does not model is the PRM problem itself. Prior
 60 graph-and-RL studies for networked operations concentrate on demand fore-
 61 casting, inventory-only ordering, or generic enterprise resource allocation,
 62 and they represent demand as fungible, abstract requests [13, 15, 18, 19,
 63 20]; none model a product that expires, production that must return to the
 64 same patient, or demand generated by patient deterioration while waiting.
 65 Reinforcement learning has been applied within cell-therapy manufacturing,
 66 but at the level of a single bioprocess rather than a network of clinics [2],
 67 and recent evidence cautions that learned policies do not uniformly domi-
 68 nate well-tuned operational heuristics, particularly under perishability and
 69 non-stationarity [21]. The gap we address is thus a problem gap rather than
 70 a method gap, namely perishable, identity-bound, patient-condition-driven
 71 capacity planning across a clinic network, together with a decision framework
 72 and simulation testbed built specifically for it. To the best of our knowledge
 73 no public benchmark exists for this setting, so the environment is itself a
 74 contribution.

75 We organize the study around five research questions. **(RQ1)** Does a
 76 graph representation of the clinic network improve coordinated capacity,
 77 reagent, and replenishment policies over flat-state control, and under which
 78 supply-disruption and forecast-error conditions is any improvement largest?
 79 **(RQ2)** Which actor-critic backbone (DDPG, TD3, SAC, or PPO) is best

80 suited to this constrained continuous-control problem, and is DDPG alone
 81 adequate? **(RQ3)** Does explicitly modeling patient condition and product
 82 expiry change the optimal policy and expose failures of condition-blind plan-
 83 ners? **(RQ4)** Does a condition- and expiry-aware temporal encoder, which
 84 retains memory of deterioration trajectories, improve on a static graph en-
 85 coder? **(RQ5)** Do learned policies outperform strong look-ahead heuristics,
 86 not only weak myopic rules, under supply disruption, demand forecast error,
 87 and patient-condition stress?

88 To study these questions we develop, first, a simulation of a multi-site au-
 89 tologous cell-therapy network in which patient health conditions and product
 90 and material expiry are first-class drivers of demand and feasibility, and sec-
 91 ond, an advantage-filtered residual GCN-DDPG (AFR-GCN-DDPG) con-
 92 troller that coordinates capacity, reagent, and replenishment decisions across
 93 the network. AFR is our shorthand for the complete controller design: a
 94 transparent MDL-2 anchor, a graph-aware bounded residual actor, evidence-
 95 filtered teacher targets, and independently calibrated deployment safeguards.
 96 It does not replace or rename DDPG. The graph component encodes each
 97 facility’s operational and patient-condition state together with information
 98 from connected facilities into network-aware embeddings. A deterministic
 99 policy-gradient actor then learns a bounded, state-dependent correction to
 100 the anchor policy, and a feasibility projection converts the corrected action
 101 into implementable controls. Advantage-filtered distillation retains a nonzero
 102 correction target only when common-random-number look-ahead predicts an
 103 improvement over the anchor; it is a teacher-data construction and regular-
 104 ization step, not a second algorithm or deployed policy. Checkpoint and

residual-scale calibration provide a deployment safeguard. DDPG is the canonical proposed instantiation and provides continuity with prior PRM capacity-planning work; graph-aware TD3, SAC, and PPO variants serve as controlled stability and backbone comparisons. We adopt a statistically disciplined evaluation protocol (multiple random seeds, paired Monte Carlo outcomes, and two-level confidence intervals), because single-seed, mean-only reporting can make deep reinforcement learning results appear stronger and more stable than they are [22, 23]. Consistent with the clinical stakes of the problem, we report patient-service outcomes such as eligibility and waiting alongside operating cost, since a lower average cost can mask clinically unacceptable delay.

The main contributions of this study are as follows.

- We formulate distributed capacity planning for *perishable, identity-bound, patient-condition-driven* PRM manufacturing as a constrained networked sequential decision problem, in which patient deterioration and product and material expiry are explicit determinants of demand, feasibility, and cost.
- We develop a simulation environment realizing this formulation for a multi-clinic autologous cell-therapy network (with patient-condition dynamics, expiry-aware inventory, and resource sharing under the autologous identity constraint) that is reusable as a benchmark testbed.
- We propose AFR-GCN-DDPG, an advantage-filtered residual graph controller that learns bounded corrections to a transparent look-ahead anchor under demand and disruption uncertainty, with persistent teacher

regularization and independently calibrated deployment safeguards.

- We establish a paired multi-seed evaluation protocol for comparisons with strong look-ahead and isolated-facility heuristics, flat-state reinforcement learning, and graph-aware backbone variants, and report a five-seed DDPG confirmation, a parameter-matched graph-attribution study, and a conservative online-TD3 screen with two-level bootstrap confidence intervals.
- We draw design implications for intelligent decision support in distributed, patient-specific healthcare manufacturing and related time-critical, make-to-order engineering systems.

The remainder of this paper is organized as follows. Section 2 formulates the distributed PRM capacity planning problem, extended with patient-condition states and product and material expiry. Section 3 reviews personalized regenerative medicine operations, reinforcement- and graph-learning for networked operational decisions, and the method foundations of continuous-control reinforcement learning and graph neural networks, closing with the design implications that motivate our approach. Section 4 presents the graph-aware deep reinforcement learning framework. Section 5 describes the computational experiments and results. Section 6 concludes the study and identifies future research directions.

Table 1: Notation used in the distributed manufacturing and capacity planning model.

Sets and parameters	
Γ	Planning horizon indexed by t .
T	Manufacturing time for one PRM product, in periods, where $T < \Gamma$.
N	Set of manufacturing facilities indexed by n .
m_t^n	Products that begin manufacturing at facility n at epoch t .
d_t^n	Demand arrivals at facility n in epoch t .
State variables	
s_t^n	Specimens waiting for manufacturing at facility n at epoch t .
r_t^n	Available reagent units at facility n at epoch t .
$\mathbf{b}_t^n$	Bioreactor work-in-process vector $(b_t^{n,1}, b_t^{n,2}, \dots, b_t^{n,T})$, where $b_t^{n,\tau}$ denotes the number of bioreactors at facility n with τ periods remaining at epoch t .
P_t^n	Maximum reagent purchase quantity for facility n for epoch $t + 1$.
Decision variables	
w_t^n	Specimens added to or subtracted from facility n at epoch t and received or removed before epoch $t + 1$; fixed to zero under the autologous identity constraint (Section 2.3).
e_t^n	Reagent units added to or subtracted from facility n at epoch t and received or removed before epoch $t + 1$.
q_t^n	Idle bioreactors added to or subtracted from facility n at epoch t and received or removed before epoch $t + 1$.
p_t^n	Reagent purchase quantity for facility n for epoch $t + 1$.
Patient-condition and perishability	
H_j	Latent health index of patient j , drawn $H_j \sim \text{Uniform}(0, 1)$ at enrollment.
τ_j	Deterioration epoch of patient j , drawn $\tau_j \sim \theta \text{Weibull}(\kappa)$.
κ, θ	Shape and scale of the Weibull deterioration-epoch distribution.
α_j	Number of epochs since patient j enrolled, including waiting and manufacturing time.
S_j	Survival index of patient j , $S_j \in [0, 1]$; the patient is eligible while $S_j \geq \epsilon$.
λ_H, λ_F	Survival decay rates of an ideally healthy and a maximally frail

149 2. Distributed Network Manufacturing and Capacity Planning Prob- 150 lem

151 In this section, we formulate the capacity planning problem for distributed
152 manufacturing systems as a production simulation model based on a Markov
153 decision process (MDP). Suppose the distributed manufacturing system con-
154 sists of N manufacturing facilities that produce one type of make-to-order
155 PRM product. Each manufacturing facility has its own production capacity,
156 including bioreactors and reagent inventory, and can purchase reagents from
157 source material suppliers. During manufacturing, facilities receive patients’
158 specimens and initiate production based on the available specimens, biore-
159 actors, and reagents. We assume that reagent transshipment and bioreactor
160 capacity relocation are allowed among facilities, whereas patient specimens
161 and finished products are identity-bound and cannot be pooled or reassigned
162 across facilities (Section 2.3). Demand arrivals and supplier disruptions are
163 modeled as stochastic processes, implying that local supply may be either
164 insufficient or excessive relative to local demand at different decision epochs.
165 The notation used throughout this section is summarized in Table 1.

166 Figure 1 illustrates the basic structure of the distributed manufacturing
167 problem using a two-facility example. Each facility maintains a patient-
168 specimen queue whose members carry a deteriorating survival index, together
169 with reagent inventory, bioreactor capacity, and perishable finished product.
170 At each decision epoch, condition-driven patient demand and disruption-
171 prone reagent replenishment arrive under uncertainty. When one facility
172 experiences excess demand, insufficient reagents, or limited production ca-
173 pacity, the network may use reagent transshipment or bioreactor capacity

174 relocation to rebalance the two *fungible* resources across facilities; patient
 175 specimens and finished product, being identity-bound, remain fixed to their
 176 enrolling facility. This structure motivates the MDP formulation, where the
 177 state captures facility-level demand, patient condition, inventory, and capac-
 178 ity, and the action determines how fungible resources should be replenished,
 179 shared, or relocated across the network.

180 For the N facilities, T epochs of manufacturing time are required to
 181 produce one of the PRM products, and the planning horizon consists of Γ
 182 epochs where $\Gamma > T$.

183 At epoch $t \in \{1, 2, \dots, \Gamma\}$, the production status of the n -th facility for
 184 $n = 1, \dots, N$ is expressed as a state vector $\mathbf{x}_t^n$, where

$$\mathbf{x}_t^n = (s_t^n, r_t^n, \mathbf{b}_t^n, P_t^n), \quad (1)$$

185 Based on the state variables at epoch t , the facilities need to take action
 186 to adjust their production capacity for epoch $t + 1$. We define the action
 187 $\mathbf{a}_t^n$ as a vector that consists of the amount of capacity sharing and reagent
 188 replenishment. The action is decided at the beginning of the production
 189 period t and executed before the start of $t + 1$.

$$\mathbf{a}_t^n = (w_t^n, e_t^n, q_t^n, p_t^n), \quad (2)$$

190 The goal of capacity planning is to minimize total production cost by
 191 dynamically selecting the actions from Equation (2) for all facilities, denoted
 192 by $\mathbf{A}_t = (\mathbf{a}_t^1, \dots, \mathbf{a}_t^N)$, for each epoch $t \in \Gamma$. We define $V(\mathbf{X}_t)$ as the total
 193 future cost of the selecting actions $\mathbf{A}_t$ when the states from all facilities,
 194 denoted by $\mathbf{X}_t = (\mathbf{x}_t^1, \dots, \mathbf{x}_t^N)$, are observed at epoch $t \in \Gamma$. The objective

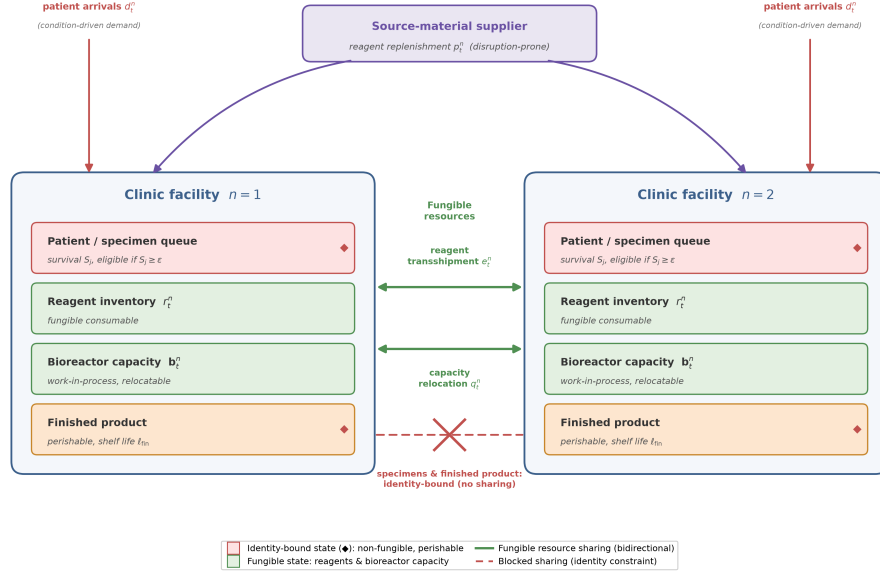


Figure 1: Illustration of a two-facility distributed manufacturing system for personalized regenerative medicine. Each facility holds four state components, namely a patient–specimen queue whose members carry a deteriorating survival index S_j (eligible while $S_j \geq \epsilon$), reagent inventory r_t^n , bioreactor capacity b_t^n , and perishable finished product. Condition-driven patient demand d_t^n and disruption-prone reagent replenishment p_t^n arrive under uncertainty. The two *fungible* resources (reagents and bioreactor capacity) can be rebalanced across facilities through transshipment e_t^n and capacity relocation q_t^n (solid green links). Patient specimens and finished product are *identity-bound*, so they cannot be pooled or reassigned across facilities (crossed red link). This fungible-only sharing under an identity constraint is the operational asymmetry that distinguishes distributed PRM manufacturing.

195 function can be expressed as a dynamic programming problem:

$$V(\mathbf{X}_t) = \min_{\mathbf{A}_t} E[c(\mathbf{A}_t; \mathbf{X}_t) + \gamma V(\mathbf{X}_{t+1})], \quad (3)$$

196 which is useful for finding the optimal action $\mathbf{A}_t$ for each state $\mathbf{X}_t$ at
 197 each epoch $t \in \Gamma$, where $c(\mathbf{A}_t; \mathbf{X}_t)$ is the immediate cost at time t , and
 198 $\gamma \in [0, 1]$ is a discount factor for balancing the impact of future costs in
 199 deciding actions. The choice of γ depends on the user’s goal; if γ is closer
 200 to 1 (to 0), then it means the user cares more about long (short) term
 201 cost; if $\gamma = 0$, then only the cost for current epoch is considered. Equation
 202 (3) generalizes the optimality equation for the capacity planning problem in
 203 Li(2021) [6] in that cost functions $\mathbf{c}$ and $\mathbf{V}$ and the cumulative distribution
 204 function for the expectation operator can be unknown.

205 Equations (1)–(3) describe the base operational MDP shared with generic
 206 networked replenishment problems. The remainder of this section extends it
 207 along the three dimensions that distinguish PRM capacity planning, namely
 208 the deteriorating clinical condition of the patients awaiting treatment, the
 209 hard expiry of biological source material and finished product, and the autol-
 210 ogous identity constraint that forbids pooling of patient-specific specimens.
 211 Section 4 then develops a graph-aware DRL approach that learns the cost
 212 function $\mathbf{c}$, the value function $\mathbf{V}$, and the expectation operator directly from
 213 simulated experience.

214 2.1. Patient condition and treatment eligibility

215 In generic capacity planning, a unit of unmet demand is deferred, back-
 216 ordered, or lost at a fixed penalty, and its value does not change while it

217 waits. PRM demand behaves differently, because each arrival is a patient
 218 whose clinical condition deteriorates throughout the vein-to-vein pathway.
 219 A specimen that waits too long, or a therapy whose matched patient deterio-
 220 rates during manufacturing, may no longer produce a deliverable treatment.
 221 We therefore attach an explicit condition process to every enrolled patient
 222 from queue entry through manufacturing completion, adapting the real-time
 223 patient-health modeling of Tseng et al. [24] to a distributed capacity-planning
 224 setting.

225 When a patient j enrolls at facility n , we draw a latent health index $H_j \sim$
 226 $\text{Uniform}(0, 1)$ and a deterioration epoch $\tau_j \sim \theta \text{Weibull}(\kappa)$ with shape κ and
 227 scale θ ; H_j captures the heterogeneous baseline robustness across patients,
 228 while τ_j marks the onset of accelerated decline. Let α_j be the number of
 229 epochs since enrollment, including both queueing and manufacturing time.
 230 The patient’s survival index $S_j \in [0, 1]$ (a normalized proxy for the likelihood
 231 that the patient remains treatable) evolves as

$$S_j(\alpha_j) = \ell(\alpha_j) + H_j(u(\alpha_j) - \ell(\alpha_j)), \quad (4)$$

232 where $u(\alpha) = \exp(-\lambda_H g(\alpha))$ and $\ell(\alpha) = \exp(-\lambda_F g(\alpha))$ are the survival
 233 trajectories of an ideally healthy and a maximally frail patient, with decay
 234 rates $\lambda_H < \lambda_F$, so that H_j interpolates each patient between the two bounds.
 235 The effective exposure time

$$g(\alpha) = \begin{cases} \alpha, & \alpha \leq \tau_j, \\ \tau_j + \rho(\alpha - \tau_j), & \alpha > \tau_j, \end{cases} \quad (5)$$

236 accelerates by a factor $\rho > 1$ once the patient passes the deterioration epoch

τ_j , reflecting the clinically observed transition to rapid decline. A patient remains eligible for treatment only while $S_j \geq \epsilon$. Before production starts, crossing the threshold removes the patient and specimen from the waiting queue. After production starts, the same condition process continues through every manufacturing epoch. If the matched patient becomes ineligible before completion, the in-process therapy is discarded and the occupied bioreactor is cleaned during that epoch before returning to the idle pool at the next decision epoch. Treatment service is recorded only when manufacturing completes and the therapy is infused, not when production starts. Because condition erodes monotonically in expectation and accelerates after τ_j , neither waiting nor manufacturing delay is a neutral buffering decision. Parameter values are reported in Section 5.

2.2. Perishability of materials and finished product

Both the biological source material and the finished product are perishable and identity-specific, so they cannot be stockpiled against future demand. A waiting specimen that is not put into production within ℓ_{mat} epochs of enrollment expires, so the patient is lost and the reserved material is wasted. Finished product that is not delivered within ℓ_{fin} epochs of completion likewise expires and is discarded. We track, per facility n and epoch t , patients lost to waiting-stage ineligibility $\phi_t^{n,\text{wait}}$, manufacturing-stage ineligibility $\phi_t^{n,\text{mfg}}$, and specimen expiry $\phi_t^{n,\text{exp}}$, together with finished units lost to expiry $\psi_t^{n,\text{fin}}$. A manufacturing-stage ineligibility also creates one discarded in-process therapy. It is convenient to collect the total patients lost and total material wasted at each facility,

$$\phi_t^n = \phi_t^{n,\text{wait}} + \phi_t^{n,\text{mfg}} + \phi_t^{n,\text{exp}}, \quad \psi_t^n = \phi_t^{n,\text{exp}} + \phi_t^{n,\text{mfg}} + \psi_t^{n,\text{fin}}, \quad (6)$$

where an expired specimen contributes both a lost patient and a unit of wasted material. To signal impending losses before they occur, we also count the *at-risk unserved* patients whose survival has fallen within an urgency margin δ of the eligibility boundary, $v_t^n = |\{j \text{ waiting at } n : S_j < \epsilon + \delta\}|$.

2.3. Autologous identity constraint and admissible transfers

The autologous nature of PRM imposes a constraint with no analog in commodity supply chains. A specimen and the product manufactured from it are bound to a single patient and cannot be substituted, pooled, or delivered to anyone else. Consequently, although the network may rebalance fungible resources (reagents and bioreactor capacity) across facilities, it can never transship specimens or finished product to serve another facility's demand. In the notation of Equation (2) this fixes the specimen-adjustment component to $w_t^n \equiv 0$ for all n and t , so the effective per-facility decision reduces to reagent transfer e_t^n , capacity relocation q_t^n , and reagent purchase p_t^n , each subject to inventory and capacity feasibility. The network therefore moves fungible capacity toward where identity-bound work already sits, rather than moving the work toward idle capacity, the defining operational asymmetry of distributed PRM manufacturing.

2.4. Extended cost and the constrained graph MDP

The immediate cost augments the operational base cost c_{op} (covering reagent purchase, holding, and shortage; bioreactor holding and shortage;

282 and reagent and capacity transfer) with three condition- and perishability-
 283 driven terms:

$$c(\mathbf{A}_t; \mathbf{X}_t) = c_{\text{op}}(\mathbf{A}_t; \mathbf{X}_t) + \omega_{\text{lost}} \sum_{n \in N} \phi_t^n + \omega_{\text{exp}} \sum_{n \in N} \psi_t^n + \omega_{\text{urg}} \sum_{n \in N} v_t^n, \quad (7)$$

284 with weights ordered $\omega_{\text{lost}} \gg \omega_{\text{exp}} > \omega_{\text{urg}}$ to encode the clinical priority, in
 285 which losing a patient dominates wasting material, which in turn dominates
 286 the shaping penalty for letting patients drift toward the eligibility boundary.
 287 The urgency term v_t^n steers the controller to act pre-emptively (starting
 288 production or relocating capacity) before losses are actually incurred.

289 For clarity, we distinguish three quantities throughout the paper. For a
 290 policy π , the expected total weighted objective is

$$J_{\text{tot}}^\pi = \mathbb{E}_\pi \left[\sum_{t=0}^{\Gamma-1} c(\mathbf{A}_t; \mathbf{X}_t) \right] = J_{\text{op}}^\pi + J_{\text{pat}}^\pi, \quad (8)$$

291 where J_{op}^π accumulates reagent purchase, holding, and shortage; bioreactor
 292 holding and shortage; and reagent and capacity transfer, whereas J_{pat}^π accu-
 293 mulates the weighted patient-loss, expiry, and urgency terms in Equation (7).
 294 The total objective is the primary optimization and evaluation criterion. The
 295 two component groups are reported separately to show which decisions gen-
 296 erate an improvement.

297 All three quantities are reported in *modeled weighted-objective units*. Resource-
 298 price coefficients can be interpreted as monetary equivalents under the adopted
 299 calibration, but the patient-loss, expiry, and urgency weights are penalty
 300 or shadow-value parameters rather than observed accounting expenditures.
 301 Consequently, an absolute difference in J_{tot} is not presented as realized cash

302 savings without an external economic calibration. Some large components
 303 are driven partly by exogenous demand and patient deterioration and by
 304 structural constraints such as finite capacity and lead times. They are there-
 305 fore only partly controllable, not “fixed” or wholly unavoidable.

306 For an anchor policy π_{anc} and candidate π , the headline relative improve-
 307 ment retains the complete objective as its denominator,

$$R_{\text{tot}} = \frac{J_{\text{tot}}^{\pi_{\text{anc}}} - J_{\text{tot}}^{\pi}}{J_{\text{tot}}^{\pi_{\text{anc}}}}. \quad (9)$$

308 We accompany it with absolute paired differences, component-specific rela-
 309 tive changes, and clinical outcomes. When a valid oracle or lower-cost refer-
 310 ence π_{ref} is available, we additionally report the fraction of anchor headroom
 311 captured,

$$R_{\text{headroom}} = \frac{J_{\text{tot}}^{\pi_{\text{anc}}} - J_{\text{tot}}^{\pi}}{J_{\text{tot}}^{\pi_{\text{anc}}} - J_{\text{tot}}^{\pi_{\text{ref}}}}. \quad (10)$$

312 This decomposition prevents a large patient-penalty or capacity-shortage
 313 component from obscuring a meaningful change in a directly affected op-
 314 erating component, while avoiding a post hoc replacement of the primary
 315 denominator.

316 To act on this cost, the controller needs visibility into the condition of
 317 each waiting queue. We therefore extend the facility state of Equation (1)
 318 with a condition summary $\boldsymbol{\sigma}_t^n$,

$$\tilde{\mathbf{x}}_t^n = (s_t^n, r_t^n, \mathbf{b}_t^n, P_t^n, \boldsymbol{\sigma}_t^n), \quad (11)$$

319 where $\boldsymbol{\sigma}_t^n$ contains the waiting count, waiting mean survival, near-expiry
 320 count, waiting survival-band histogram, in-production count, in-production
 321 mean survival, and in-production at-risk count. The controller can therefore

anticipate both queue losses and therapies at risk of discard rather than merely react to aggregate demand.

Finally, we cast the extended problem as a *constrained graph MDP*. The facilities and their admissible reagent- and capacity-sharing links form a graph $\mathcal{G} = (N, \mathcal{E})$; specimen and finished-product edges carry no flow under the identity constraint, so only fungible-resource edges are active. The global state $\tilde{\mathbf{X}}_t = (\tilde{\mathbf{x}}_t^1, \dots, \tilde{\mathbf{x}}_t^N)$ is a signal over the nodes of $\mathcal{G}$, and the action $\mathbf{A}_t$ assigns per-node purchases and per-edge transfers subject to $w_t^n \equiv 0$ and to inventory and capacity feasibility. The optimality equation (3) carries over verbatim with the extended state $\tilde{\mathbf{X}}_t$ and cost (7), but the value function must now trade current operating cost against the future risk of patient losses and expiry. This constrained graph MDP (perishable, identity-bound, and patient-condition-driven) is the problem the remainder of the paper addresses, and its graph structure is precisely what the method of Section 4 is designed to exploit.

3. Literature Review

We review three bodies of work that our approach draws together, namely the operations of personalized regenerative medicine (PRM) and decentralized manufacturing; reinforcement learning and graph learning for networked operational decisions; and the underlying method foundations of continuous-control reinforcement learning and graph neural networks. Rather than surveying each exhaustively, we read them as an argument in which each stream supplies part of what a distributed PRM capacity-planning controller needs, none supplies all of it, and together they imply a small set of design choices

that we make explicit at the end of the section. Throughout, we treat the learning machinery as established and concede that graph representation and reinforcement learning have already been combined for supply-chain problems; what remains open is the PRM problem itself, one that is perishable, identity-bound, and patient-condition-driven.

3.1. *Personalized regenerative medicine and decentralized manufacturing*

PRM supply chains differ substantially from conventional make-to-stock manufacturing systems because products are individualized, time-sensitive, and tightly coupled with patient-specific clinical pathways. In autologous cell therapy, patient material must be collected, transported, manufactured, quality-tested, and returned for administration, and manufacturing can begin only when a specimen, consumables, and production capacity are jointly available. These characteristics make fulfillment time, service reliability, and resource utilization central operational measures. Wang et al. [1] used simulation to compare centralized and point-of-care autologous cell therapy supply chain strategies and evaluated cost, fulfillment time, service level, and resource utilization. Li and White [6] further studied decentralized autologous cell therapy manufacturing networks and showed that transshipment and relocatable capacity can reduce the cost of resilience under supplier disruption. Most directly relevant to our patient-condition modeling, Tseng et al. [24] integrated real-time patient health data into an autologous cell-therapy simulation and showed that representing patient condition, rather than treating demand as static, changes fulfillment and eligibility outcomes. We adopt this patient-condition perspective as a first-class driver of demand and feasibility, but embed it within a networked capacity-planning formulation solved

371 by graph-aware learned control, rather than evaluating fixed policies in a
372 simulation study alone.

373 These studies establish that PRM network design and inter-facility co-
374 ordination are first-order engineering decisions rather than implementation
375 details. Related work on cell therapy manufacturing process control also
376 highlights the potential of reinforcement learning in uncertain, nonlinear,
377 and partially observed manufacturing environments [2]. Together, this liter-
378 ature motivates a data-driven decision-support framework that can represent
379 both the temporal dynamics of patient-specific manufacturing and the spatial
380 dependencies created by distributed facilities.

381 *3.2. Reinforcement learning for supply chain and manufacturing decisions*

382 Reinforcement learning has been widely studied for sequential supply
383 chain and manufacturing decisions. Early work applied RL to inventory
384 replenishment in multi-echelon supply chains and perishable inventory sys-
385 tems [25, 26]. More recent studies have used deep reinforcement learning
386 to handle high-dimensional state spaces, stochastic demand, and simulation-
387 based decision environments. For example, Oroojlooyjadid et al. [27] devel-
388 oped a deep Q-network approach for the beer game, demonstrating that DRL
389 can learn replenishment policies in decentralized, partially observed supply
390 chain settings.

391 Recent supply chain RL studies have moved toward more realistic con-
392 straints and engineering settings. Bermudez et al. [19] proposed distribu-
393 tional constrained reinforcement learning for multi-period supply chain op-
394 timization with production and inventory constraints. Eisenach et al. [20]
395 studied shared-resource capacity control in inventory management using a

neural coordination mechanism and a reinforcement learning buying policy. Stranieri et al. [21] benchmarked classical and deep reinforcement learning inventory policies for pharmaceutical supply chains with perishability and non-stationary demand. These studies support the use of simulation-trained RL policies for uncertain operational control, but most focus on replenishment or inventory management rather than the joint capacity, reagent, and replenishment decisions (under a perishable, identity-bound demand process) that arise in distributed PRM manufacturing.

3.3. *Structure-informed, residual, and baseline-relative reinforcement learning*

Inventory-control evidence does not support treating generic end-to-end DRL as automatically superior to established operating rules. Gijsbrechts et al. [28] found that carefully tuned DRL can match state-of-the-art policies across lost-sales, dual-sourcing, and multi-echelon problems, while the invited roadmap of Boute et al. [29] argues that learning methods should exploit the structural policy knowledge accumulated in inventory research. De Moor et al. [30] made this principle operational by using strong perishable-inventory heuristics as teacher policies for reward shaping, improving training speed, stability, and, in suitable instances, performance relative to the teacher. Hybrid decomposition offers another route: Stranieri et al. [31] assigned production decisions to DRL and shipping decisions to stochastic programming, obtaining more reliable performance than pure DRL. Non-stationary-demand experiments further show the value of conditioning policies on forecast information rather than assuming a fixed arrival law [32].

Residual reinforcement learning provides the most direct methodologi-

cal foundation for our controller. The residual formulation decomposes the executed action into a competent base-controller action and a learned correction, reducing the part of the control law that must be discovered from experience [33]. Staessens et al. [34] showed in an industrial-control setting that bounding the correction relative to a conventional controller can preserve its stabilizing role while adapting to uncertain operating conditions. Baseline-relative policy-improvement methods likewise motivate accepting a learned update only when there is credible evidence of improvement over the incumbent policy [35]. In stochastic inventory control, Deep Controlled Learning combines approximate policy iteration with common random numbers and statistically efficient candidate selection to overcome the difficulty generic DRL has in beating strong heuristics [36]. These ideas motivate our MDL-2 anchor, bounded graph residual, common-random-number advantage filtering, and held-out deployment calibration. Our validation fallback is an empirical deployment safeguard, however, and should not be interpreted as inheriting the formal monotonic-improvement or closed-loop stability guarantees proved under the assumptions of those methods.

3.4. *Deep reinforcement learning for regenerative medicine capacity planning*

The closest antecedent to this work is Tseng et al. [7], who developed a deep reinforcement learning approach for dynamic capacity planning in decentralized regenerative medicine supply chains. Their study demonstrated that simulation-trained policies can adapt to demand and supply uncertainty without requiring exact prior knowledge of the underlying stochastic processes. This is directly aligned with the objective of using data-driven policy learning for PRM capacity planning.

446 The present study builds on this line of work by adding an explicit graph
 447 representation layer to encode inter-facility relationships. In decentralized
 448 PRM networks, facilities are linked through reagent transshipment and biore-
 449 actor capacity relocation, whereas patient specimens and finished product
 450 remain identity-bound. Therefore, a facility’s decision depends not only on
 451 its own local state but also on the states and fungible-resource conditions
 452 of connected facilities. A flat vectorized state representation can include
 453 all facility variables, but it does not explicitly exploit the graph structure
 454 that governs how information and resources propagate across the network.
 455 This motivates integrating graph convolutional representation learning with
 456 continuous-control actor–critic reinforcement learning.

457 *3.5. Multi-agent reinforcement learning and graph learning for networked* 458 *supply chains*

459 Recent multi-agent reinforcement learning studies are relevant because
 460 distributed manufacturing networks naturally involve decentralized infor-
 461 mation and coordinated actions. Mousa et al. [18] formulated decentral-
 462 ized inventory control as a multi-agent problem and studied centralized-
 463 training/decentralized-execution policies under local information constraints.
 464 Leluc et al. [37] similarly demonstrated the promise of cooperative multi-
 465 agent reinforcement learning for inventory management under stochastic de-
 466 mand and lead times. These studies show that decentralization is not only
 467 an application feature but also an algorithmic design dimension.

468 In parallel, graph neural networks have emerged as a natural repre-
 469 sentation tool for supply chains and other networked operational systems.
 470 Chang et al. [14] learned production functions on supply-chain graphs us-

471 ing temporal graph neural networks and an inventory module, showing the
 472 value of explicit relational structure for supply-chain inference and fore-
 473 casting. Ahn et al. [15] used attention-based graph neural networks for
 474 probabilistic supply and inventory prediction in enterprise supply-chain net-
 475 works, while Ahn et al. [16] combined graph neural networks, offline re-
 476 inforcement learning, and policy simulation for dynamic supply planning.
 477 Kotecha and del Rio Chanona [13] provide the closest graph-control an-
 478 tecedent: their GNN-MARL framework dynamically parameterizes heuris-
 479 tic inventory policies and uses graph structure to coordinate decentralized
 480 supply-chain agents. AFR-GCN-DDPG differs by learning a bounded con-
 481 tinuous correction around a fixed look-ahead anchor, filtering correction tar-
 482 gets through common-random-number look-ahead, and controlling patient-
 483 condition, capacity, and fungible-resource decisions in an identity-bound
 484 PRM network rather than inventory replenishment alone. These studies
 485 collectively suggest that graph-based learning is useful for networked sup-
 486 ply chain systems; however, most existing work remains focused on predic-
 487 tion, inventory-only control, or general supply-chain planning rather than
 488 continuous-action control of PRM-specific capacity, reagent, and specimen
 489 flows.

490 *3.6. Method foundations: continuous-control reinforcement learning and graph* 491 *neural networks*

492 The decisions in our problem (capacity relocation, reagent replenishment,
 493 and consumable transfer) are real-valued, which points to continuous-control
 494 actor-critic methods. Deterministic policy gradients and DDPG [8] extended
 495 actor-critic learning to continuous action spaces, but are known to suffer from

value overestimation and training instability. TD3 [9] mitigates overestimation through clipped double- Q learning, delayed policy updates, and target smoothing; SAC [10] adds entropy regularization for more robust exploration and stability; and PPO [11] is a widely used on-policy alternative built on a clipped surrogate objective. The practical implication is that DDPG, though historically the default for continuous supply-chain control, is no longer an uncontested choice. A defensible study should compare backbones rather than commit to one, and should expect the more stable off-policy methods to reduce the run-to-run variance that plagues DDPG.

Graph neural networks encode relational structure through neighborhood aggregation. The graph convolutional network [12] and its successors (GraphSAGE [38] for inductive neighborhood sampling and graph attention networks [39] for learned edge weighting) provide a permutation-equivariant inductive bias that a flat multilayer perceptron lacks. Coupling such encoders with reinforcement learning has become a recognized paradigm for networked decision problems [17]. For a clinic network whose facilities interact through capacity relocation and consumable transshipment, this relational bias is a modeling choice of substance rather than convenience, because it allows a single policy to reason over the topology along which shortages, slack, and disruptions propagate.

3.7. *Evaluation considerations for deep reinforcement learning*

Evaluation rigor is especially important for DRL-based engineering applications. Henderson et al. [22] showed that deep reinforcement learning results can be sensitive to random seeds, hyperparameters, and implementation details. Agarwal et al. [23] further argued that few-run comparisons can

521 be statistically unreliable and recommended more robust aggregate metrics
522 and interval estimates. These findings imply that a DRL-based supply chain
523 study should report implementation details, training budgets, out-of-sample
524 test procedures, and variability across replications or random seeds when
525 possible. In the present study, we therefore emphasize common simulation
526 settings across all benchmark policies, detailed implementation parameters
527 for the graph-aware actor-critic agents, and Monte Carlo evaluation under
528 multiple disruption and demand forecast scenarios, with interquartile-mean
529 performance and confidence intervals reported across random seeds.

Table 2: Positioning of this study relative to closely related literature.

Literature stream	Representative studies	Main limitation relative to this work
PRM operations	Simulation and decentralized planning [1, 6, 7]	No learned graph policy combining patient condition, perishability, identity, and coordinated continuous control.
Supply-chain RL	Inventory and constrained control [27, 19, 21, 20]	Primarily inventory or replenishment, not coupled PRM capacity and fungible-resource sharing.
Decentralized MARL	Cooperative inventory control [18, 37]	Not tailored to identity-bound, patient-specific make-to-order manufacturing.
Supply-chain GNNs	Graph forecasting and planning [14, 15, 16, 13]	Mostly predictive or inventory-focused rather than continuous PRM network control.
Structured residual RL	Heuristic-guided and baseline-relative learning [30, 34, 35, 36]	No graph-structured PRM application with patient-condition risk and identity constraints.

530 3.8. *Research gap and design implications*

531 Read together, these literatures imply five design choices for a distributed
532 PRM capacity-planning controller. First, because the decisions are sequen-
533 tial and real-valued, the controller should use continuous-control actor–critic
534 reinforcement learning rather than value-based or discretized methods. Sec-
535 ond, because a facility’s value depends on its network position and resources
536 propagate along inter-facility links, its state should be encoded with a graph
537 rather than a flat vector. Third, because strong operational rules already
538 solve much of the nominal problem, the learned policy should target bounded,
539 evidence-supported corrections to a transparent anchor rather than relearn
540 the complete policy from scratch. Fourth, DDPG provides a natural canon-
541 ical formulation and continuity with earlier PRM capacity-planning work,
542 but its known instability requires controlled comparisons with TD3 and sec-
543 ondary SAC/PPO variants rather than an untested assumption that DDPG
544 is always best. Fifth, because reinforcement learning results are sensitive
545 to seeds and implementation details, evaluation should report multi-seed in-
546 terquartile means with confidence intervals and, given the clinical stakes,
547 patient-service outcomes such as eligibility and waiting.

548 None of these streams, however, addresses the problem that defines our
549 setting. PRM operations research motivates adaptive, networked capacity
550 planning but does not learn relational policies. Supply-chain graph-and-
551 reinforcement-learning research supplies the relational-control machinery but
552 treats demand as fungible and abstract, targeting forecasting or inventory-
553 only control. Cell-therapy reinforcement learning operates on a single biopro-
554 cess rather than a clinic network. As a result, no prior work models the three

555 defining properties of PRM capacity planning (a perishable product with a
 556 hard viability deadline, an identity-bound product that cannot be pooled
 557 or substituted across patients, and demand driven by patient deterioration
 558 while waiting) within a single graph-aware continuous-control formulation.
 559 Our contribution closes this gap. We cast the problem as a constrained
 560 graph MDP, build a reusable simulation testbed that realizes it, and study a
 561 family of graph-aware actor–critic controllers, pairing a GCN state encoder
 562 with DDPG, TD3, SAC, and PPO, against strong flat-state and heuristic
 563 baselines.

564 **4. Graph-Aware Deep Reinforcement Learning Framework**

565 *Overview: a shared graph encoder with interchangeable actor–critic backbones*

566 Distributed manufacturing networks for personalized regenerative medicine
 567 (PRM) comprise complex, interconnected facilities whose operating states
 568 are best represented as graphs rather than flat vectors. Classical reinforce-
 569 ment learning that treats each facility as independent cannot capture these
 570 spatial dependencies. We therefore adopt a two-part design shared by ev-
 571 ery method we study. A graph convolutional network (GCN) encoder maps
 572 the networked state to permutation-equivariant, neighborhood-aware node
 573 embeddings, composed with a continuous-control actor–critic backbone that
 574 maps those embeddings to feasibility-projected capacity, reagent, and re-
 575 plenishment actions. Writing the controller as *encoder* $\circ$ *backbone* lets us
 576 hold the relational representation fixed while swapping the reinforcement-
 577 learning backbone, so that any performance difference between backbones is
 578 attributable to the learning algorithm rather than to the state representation.

579 *Primary AFR-GCN-DDPG controller and backbone variants*

580 The primary proposed controller is advantage-filtered residual GCN-DDPG,
581 abbreviated AFR-GCN-DDPG. The acronym names the complete controller
582 rather than a new actor-critic backbone: *residual* means that the DDPG
583 actor modifies, rather than replaces, the MDL-2 anchor; *advantage-filtered*
584 means that nonzero teacher corrections are retained only when paired common-
585 random-number look-ahead estimates a positive improvement over that an-
586 chor. This filtering is a teacher-data construction and regularization step
587 inside AFR-GCN-DDPG, not a separate deployed policy. The controller
588 combines the relational inductive bias of a GCN with the deterministic
589 continuous-control mechanism used in earlier PRM capacity-planning re-
590 search, while reducing the burden of relearning a complete operational policy
591 from scratch. Following residual and constrained-residual control [33, 34], the
592 actor learns only a bounded correction to a transparent look-ahead anchor.
593 We also evaluate TD3, SAC, and PPO behind the same GCN encoder (Ta-
594 ble 4) as controlled backbone variants: TD3 through clipped double- Q learn-
595 ing, delayed policy updates, and target smoothing; SAC through entropy-
596 regularized stochastic policies; and PPO through an on-policy clipped surro-
597 gate objective. All four share the graph representation, feasibility-projection
598 layer, environment, and evaluation protocol, so backbone comparisons do
599 not erase AFR-GCN-DDPG’s role as the canonical proposed instantiation.
600 A GCN encodes relational structure but has no temporal memory of its own;
601 the actor-critic component supplies the sequential decision dynamics, and
602 the two together encode spatial correlations and temporal trade-offs jointly.
603 We therefore present the proposed AFR-GCN-DDPG controller in full

Table 3: Terminology for the proposed controller and its matched comparisons.

Term	Meaning and role in this study
GCN; DDPG; TD3; SAC; PPO	Established acronyms. Standard graph-encoding and actor-critic algorithms, used with their conventional meanings and cited source formulations.
MDL-2	Study baseline. Two-period mean-demand look-ahead heuristic and transparent anchor; the label is not a general reinforcement-learning acronym.
Advantage-filtered distillation	Training step. Common-random-number look-ahead retains a nonzero teacher correction only when it improves on the anchor. It is not a separate policy.
AFR	Study-specific prefix. Shorthand for the complete <i>advantage-filtered residual</i> design: anchor, bounded learned correction, teacher regularization, feasibility projection, and deployment calibration.
AFR-GCN-DDPG	Proposed controller. AFR with a GCN state encoder and DDPG actor-critic backbone.
AFR-Flat-DDPG; AFR-GCN-TD3	Matched comparisons. The first removes the graph encoder; the second changes the RL backbone. Neither is a separate proposed framework.

Table 4: Continuous-control actor-critic backbones evaluated behind the shared GCN encoder.

Backbone	Policy class	Sampling	Key mechanism and role in this study
DDPG [8]	Deterministic	Off-policy	Canonical backbone of the proposed AFR-GCN-DDPG controller; retains continuity with prior PRM capacity-planning work.
TD3 [9]	Deterministic	Off-policy	Stability-enhanced variant using clipped double- Q learning, delayed policy updates, and target smoothing.
SAC [10]	Stochastic	Off-policy	Entropy-regularized objective for more robust exploration and training stability.
PPO [11]	Stochastic	On-policy	Clipped surrogate objective; an on-policy alternative to the off-policy backbones.

604 (the anchor-plus-residual action, GCN encoder, actor-critic updates, advantage-
 605 filtered supervision, and feasibility projection) and then note where each
 606 stability-oriented variant modifies the critic target, policy parameterization,
 607 or update schedule; the graph encoder and feasibility-projection layer are
 608 identical across backbones.

609 Figure 2 shows the graph abstraction used by the proposed graph-aware
 610 framework. Each healthcare clinic is represented as a node with facility-
 611 specific state features, including waiting specimens and their condition sum-
 612 mary, reagent inventory, available bioreactor capacity, and other local op-
 613 erational conditions. The edges encode different forms of inter-facility de-
 614 pendency. Information-flow edges let the model propagate operational and
 615 patient-condition signals among neighboring clinics; resource-sharing edges
 616 represent feasible reagent transshipment relationships; and capacity-sharing
 617 edges connect each clinic to the central capacity hub. Consistent with the
 618 autologous identity constraint, only fungible reagents and bioreactor capacity
 619 flow along edges, while specimens and finished product remain node-local. By
 620 applying graph convolution over this structure, the model learns a network-
 621 aware representation in which each facility’s decision is informed not only by
 622 its own state but also by the states of connected facilities. This graph em-
 623 bedding is then passed to the actor-critic backbone to support coordinated
 624 capacity, reagent, and replenishment decisions.

625 Formally, let $G_t = (V, E_t)$ represent the manufacturing network at epoch
 626 t , where each node $v_i \in V$ denotes a facility with state features x_i^t and edge
 627 weights e_{ij}^t indicate logistic or transshipment costs. The GCN encodes local

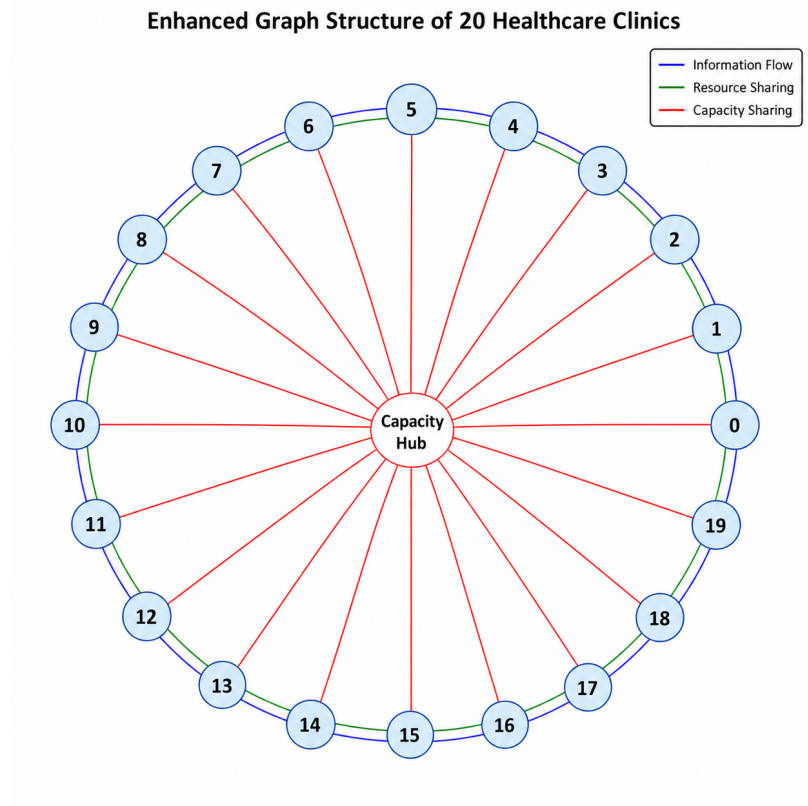


Figure 2: Graph representation of a distributed personalized regenerative medicine manufacturing network with 20 healthcare clinics. Each outer node represents a clinic or manufacturing facility, and the central node represents the shared capacity pool. Blue edges indicate information flow among neighboring clinics, green edges indicate resource sharing relationships, and red radial edges indicate capacity sharing between each clinic and the central capacity hub. This graph structure is used by the graph convolutional network to encode both local facility states and neighboring network information for downstream reinforcement learning decisions.

628 and neighborhood information using a layer-wise propagation rule:

$$h_i^{(l+1)} = \sigma\left(W^{(l)} \sum_{j \in \mathcal{N}(i)} \frac{1}{c_{ij}} h_j^{(l)}\right), \quad (12)$$

629 where $h_i^{(0)} = x_i^t$, $W^{(l)}$ are trainable weights, $\sigma(\cdot)$ is a ReLU activation, and
 630 c_{ij} is a degree-based normalization factor. The resulting embedding $H_t =$
 631 $\{h_i^{(L)}\}_{i=1}^N$ captures the current global state of the network.

632 *Overview of the graph-aware actor-critic workflow (DDPG instantiation)*

633 The method initiates with a dual initialization of the actor and critic
 634 networks of the backbone and the GCN for graph state processing. Given
 635 an observed state X_t , represented as raw graph data reflecting the current
 636 status of the manufacturing network, the GCN is used as a preprocessing
 637 unit. It transforms this raw graph into a condensed representation X'_t , which
 638 retains the critical structural information of the original graph.

639 Following this transformation, the DDPG algorithm gets involved. Using
 640 the graph-processed state X'_t , the actor-network selects an action, which is
 641 based on the actor-critic architecture. This action could range from decisions
 642 regarding resource allocation to inventory replenishments. Once executed
 643 within the manufacturing environment, the system provides a feedback loop
 644 in the form of a reward, essentially evaluating the decision's efficacy.

645 The DDPG then uses this feedback, in tandem with the replay buffer,
 646 to refine both the actor and critic networks, ensuring better decision-making
 647 in subsequent iterations. The GCN, on its part, continuously updates its
 648 understanding of the graph structure as the manufacturing network evolves,
 649 ensuring it remains adaptive and accurate in its representations.

650 *Initialization*

- 651 • Critic network $Q(X_t, A_t|\theta_Q)$ initialized with weights θ_Q .
- 652 • Actor network $\mu(X_t|\theta_\mu)$ initialized with weights θ_μ .
- 653 • GCN initialized with weights ω for processing graph states.

654 *DDPG Component*

655 The DDPG agent operates on the GCN-encoded state H_t to generate
 656 a bounded residual action $\Delta A_t = \mu(H_t|\theta_\mu)$. Let A_t^{anc} denote the action
 657 produced by the transparent look-ahead anchor and let ρ be a vector of
 658 decision-group-specific trust-region scales. The raw proposed action is

$$A_t^{\text{raw}} = A_t^{\text{anc}} + \rho \odot \Delta A_t. \quad (13)$$

This residual parameterization preserves the anchor’s basic inventory logic while allowing the GCN policy to adapt replenishment and, where enabled, fungible resource-sharing decisions to network-wide patient pressure, geographic connectivity, disruptions, and demand-prior drift. The critic network $Q(H_t, A_t|\theta_Q)$ estimates the expected cumulative cost. The optimization follows the standard DDPG updates:

$$y_i = c_i(A_i; H_i) + \gamma Q'(H_{i+1}, \mu'(H_{i+1}|\theta_{\mu'})|\theta_{Q'}), \quad (14)$$

$$L_Q = \frac{1}{N} \sum_i (y_i - Q(H_i, A_i|\theta_Q))^2, \quad (15)$$

$$\nabla_{\theta_\mu} J = \frac{1}{N} \sum_i \nabla_a Q(H_i, a|\theta_Q)|_{a=\mu(H_i)} \nabla_{\theta_\mu} \mu(H_i|\theta_\mu). \quad (16)$$

659 *Advantage-filtered residual initialization and deployment*

660 The useful residual corrections in AFR-GCN-DDPG are sparse because
661 the MDL-2 anchor is already competitive in many states. We therefore ini-
662 tialize the residual actor through advantage-filtered teacher construction. At
663 a sampled state, a small set of replenishment-only residual candidates is com-
664 pared with the zero-residual MDL-2 action over a finite look-ahead horizon.
665 Candidate and anchor rollouts use common random numbers so that their
666 score difference is not dominated by different demand, deterioration, or dis-
667 ruption realizations. A nonzero target is retained only when it improves the
668 configured operating-cost and patient-outcome score; otherwise the target is
669 the zero residual. To keep the sparse improvement labels from being over-
670 whelmed, improving and zero-residual examples receive equal total weight.
671 The weighted advantage-filtered data remain in the actor loss during DDPG
672 fine-tuning rather than being used only as a one-time warm start.

673 Deployment uses two independent validation stages. First, checkpoints
674 saved during training are compared on common validation scenarios. Second,
675 the selected checkpoint is evaluated over a fixed set of residual trust-region
676 scales, including zero. The nonzero residual is deployed only when it improves
677 validation cost without reducing service; otherwise the controller falls back to
678 MDL-2. This mechanism keeps the deployed controller within the residual
679 GCN-DDPG family while preventing a poorly generalized correction from
680 overriding the operational anchor.

681 *Reward function*

682 The agents optimize the negative of the immediate cost defined in Sec-
 683 tion 2.4. For each epoch t , the reward is

$$r_t = -c(\mathbf{A}_t; \tilde{\mathbf{X}}_t), \quad (17)$$

684 where $c(\cdot)$ is the extended cost of Equation (7), comprising the operational
 685 base cost c_{op} (reagent purchase, holding, and shortage; bioreactor holding and
 686 shortage; and reagent and capacity transfer) together with the patient-loss,
 687 material-and-product-expiry, and urgency terms. Optimizing this reward
 688 therefore requires the agent to trade current operating cost against the future
 689 risk of patient losses and expiry, rather than resource utilization alone.

690 *Action decoding and feasibility projection..* Because the actor network out-
 691 puts continuous residuals, an additional deterministic decoding and feasibility-
 692 projection step is used before actions are executed in the simulation envi-
 693 ronment. The raw anchor-plus-residual action A_t^{raw} is first scaled to the
 694 admissible range of each decision variable and then clipped to its variable-
 695 specific lower and upper bounds. Under the autologous identity constraint
 696 the specimen-adjustment component is fixed to zero, so the decoded decision
 697 comprises only the fungible quantities (reagent transshipment, bioreactor re-
 698 location, and replenishment), which are then rounded to the nearest feasible
 699 integer values.

700 The decoded action is subsequently passed through a repair operator that
 701 enforces the operational constraints of the manufacturing network. First,
 702 nonnegativity constraints are imposed so that the resulting action cannot
 703 remove more reagents or idle bioreactors from a facility than are available at

Algorithm 1 Graph-aware actor–critic for distributed PRM capacity planning (DDPG instantiation; TD3/SAC/PPO swap the critic target, policy parameterization, or update schedule).

Require: Anchor policy π_{anc} , residual candidates $\mathcal{D}$, look-ahead horizon L , discount factor γ , soft-update rate τ , actor and critic learning rates α_μ, α_Q , exploration scale σ , minibatch size N .

- 1: Initialize GCN encoder g_ω , actor $\mu(\cdot \mid \theta_\mu)$, critic $Q(\cdot, \cdot \mid \theta_Q)$, target networks μ' and Q' , and replay buffer $\mathcal{R}$.
 - 2: Build weighted teacher data by comparing $\pi_{\text{anc}}(X) + \delta$, $\delta \in \mathcal{D}$, with $\pi_{\text{anc}}(X)$ under common-random-number L -step rollouts; use target δ^* only when it improves the score and zero otherwise.
 - 3: Fit the residual actor to the weighted advantage-filtered targets and retain this loss as an actor regularizer.
 - 4: **for** each environment step t **do**
 - 5: Observe raw graph state X_t and compute graph embedding $Z_t = g_\omega(X_t)$.
 - 6: Select exploratory residual action $A_t = \pi_{\text{anc}}(X_t) + \rho \odot \mu(Z_t \mid \theta_\mu) + \epsilon_t$, where $\epsilon_t \sim \mathcal{N}(0, \sigma^2 I)$.
 - 7: Project or clip A_t to the feasible set and execute the resulting action $\hat{A}_t$.
 - 8: Observe cost/reward r_t , next state X_{t+1} , and store $(X_t, \hat{A}_t, r_t, X_{t+1})$ in $\mathcal{R}$.
 - 9: **if** an update is scheduled **then**
 - 10: Sample minibatch $(X_i, A_i, r_i, X_{i+1})_{i=1}^N$ from $\mathcal{R}$ and compute $Z_i = g_\omega(X_i)$, $Z_{i+1} = g_\omega(X_{i+1})$.
 - 11: Compute targets $y_i = r_i + \gamma Q'(Z_{i+1}, \mu'(Z_{i+1} \mid \theta_{\mu'}) \mid \theta_{Q'})$.
 - 12: Update critic by minimizing $L_Q = N^{-1} \sum_i (y_i - Q(Z_i, A_i \mid \theta_Q))^2$.
 - 13: Update actor using the deterministic policy gradient plus the weighted advantage-filtered regularizer.
 - 14: Backpropagate through the GCN encoder and softly update target networks.
 - 15: **end if**
 - 16: **end for**
 - 17: Select checkpoint and residual scale on independent validation scenarios; return the calibrated residual policy or the MDL-2 fallback.
-

704 that epoch. Second, replenishment quantities are capped by the maximum
705 purchasable reagent quantity P_t^n and by supplier availability. Third, net-
706 work conservation constraints are enforced for the fungible sharing decisions,
707 so that the total quantities of reagents and relocatable bioreactors sent out
708 from facilities must equal the total amount received by connected facilities,
709 except for external reagent purchases; specimens and finished product are
710 identity-bound and are never transferred. Finally, transshipment and relo-
711 cation actions are restricted to feasible graph edges; if no edge connects two
712 facilities, the corresponding flow is set to zero.

713 Equivalently, the executed action can be interpreted as the feasible action
714 $\hat{A}_t$ obtained by projecting the decoded actor output onto the feasible action
715 set $\mathcal{F}(X_t)$:

$$\hat{A}_t = \Pi_{\mathcal{F}(X_t)}(A_t^{\text{raw}}), \quad (18)$$

716 where $\mathcal{F}(X_t)$ is determined by current facility states, inventory availability,
717 capacity availability, supplier limits, the autologous identity constraint, and
718 graph connectivity. This projection step improves engineering credibility by
719 ensuring that all actions executed in the simulation correspond to imple-
720 mentable capacity, replenishment, and transshipment decisions.

721 Each stability-oriented backbone reuses this encoder and projection layer
722 unchanged and alters only the reinforcement-learning update. TD3 replaces
723 the single critic with clipped double- Q targets, delays actor updates, and
724 adds target-policy smoothing; SAC replaces the deterministic actor with an
725 entropy-regularized stochastic policy and a soft value target; and PPO re-
726 places off-policy replay with on-policy rollouts optimized under a clipped
727 surrogate objective. In all cases the actor and critic operate on the GCN

embedding Z_t rather than on the flat state, and the executed action is the feasibility-projected $\hat{A}_t$.

Summary

The proposed method is therefore a residual GCN-DDPG controller: a GCN encoder supplies a permutation-equivariant, network-aware state representation; a deterministic actor learns bounded corrections to a transparent look-ahead anchor; and a feasibility-projection layer guarantees that every executed action respects inventory, capacity, supplier, identity, and connectivity constraints. The final comparison first holds the anchor, residual action space, advantage-filtered teacher, encoder, projection, training budget, and evaluation seeds fixed while replacing DDPG with TD3. Residual SAC and PPO are secondary family extensions because their stochastic residual distributions and likelihood calculations require additional matched implementation work. These variants test backbone robustness without redefining them as separate proposed methods.

5. Computational Experiments

This section reports three empirical stages designed to separate anchor improvement, graph representation, and online reinforcement-learning attribution. The first stage calibrates patient-condition dynamics and evaluates replenishment residuals under demand-prior drift. The second expands the action space to geographically structured reagent and idle-capacity transfers and compares the graph policy with a parameter-matched flat policy. The third holds the pretrained graph controller fixed while testing whether conservative online TD3 updates add value beyond frozen pretraining. The exper-

752 iments compare the learned graph-aware correction policy with its heuristic
753 anchor, other heuristic and look-ahead policies, flat-state residual policies,
754 and graph-aware backbone variants under the same simulation environment.

755 5.1. Objective

756 The objective of this study is to compare the proposed residual GCN–
757 DDPG framework with traditional heuristic methods, its MDL-2 anchor, flat-
758 state residual DDPG, and stability-oriented graph-aware actor–critic variants
759 in distributed manufacturing.

760 5.2. Methodology

761 5.2.1. Benchmarking Setup

762 The final benchmark is organized as a hierarchy of matched comparisons
763 rather than a single undifferentiated algorithm ranking (Table 5). The first
764 comparison tests whether reinforcement learning improves the MDL-2 an-
765 chor. The second isolates the graph representation by replacing the GCN
766 with a flat MLP while keeping the entire residual-learning recipe fixed. The
767 third tests whether the result is specific to DDPG by repeating the residual
768 comparison with TD3. Online rollout shields and the supervised GCN shield
769 selector are reported separately because the former uses additional simula-
770 tion at decision time and the latter is not an RL policy. Pure end-to-end RL
771 and the stochastic SAC/PPO family are reported as secondary comparators
772 rather than being allowed to substitute for these mandatory tests.

773 The heuristic baselines are:

- 774 • **MYO (Myopic):** This heuristic minimizes single-period cost without
775 considering future demand.

- 776 • **MDL-1 (One-period Mean Demand Lookahead):** MDL-1 ex-
777 tends the planning horizon to one lookahead period using deterministic
778 state transitions based on mean demand.
- 779 • **MDL-2 (Two-period Mean Demand Lookahead):** MDL-2 ex-
780 tends the planning horizon to two lookahead periods using determinis-
781 tic state transitions based on mean demand.
- 782 • **ISO (Isolated Regional Facilities):** ISO assumes no resource shar-
783 ing between facilities and focuses only on local replenishment decisions.
- 784 • **F-MYO (Forecast Myopic):** F-MYO augments current-period net-
785 work balancing with the demand forecast available at the decision
786 epoch.
- 787 • **uMYO and pMYO (Patient-aware myopic policies):** uMYO
788 adds urgency-based replenishment and capacity surges, while pMYO
789 uses a bounded patient-risk workload uplift. These controls prevent
790 the proposed method from being compared only with condition-blind
791 heuristics.

792 The learning-based comparisons are:

- 793 • **MLP residual DDPG / flat-state residual DDPG:** This base-
794 line uses the same MDL-2 anchor, residual action space, DDPG actor-
795 critic structure, training budget, and feasibility projection as the pro-
796 posed method, but removes the GCN encoder. Instead, all facility-
797 level states are concatenated into a flat vector and passed directly into

798 multilayer perceptron actor and critic networks. This baseline is de-
 799 signed to answer whether performance improvements come from rein-
 800 forcement learning alone or from explicitly encoding graph-structured
 801 inter-facility dependencies.

- 802 • **Residual GCN–TD3 and matched flat residual TD3:** These poli-
 803 cies keep the anchor, advantage-filtered supervision, action constraints,
 804 training data, and evaluation streams fixed while replacing the DDPG
 805 update with TD3. They test whether twin critics, delayed actor up-
 806 dates, and target smoothing improve stability.

- 807 • **Pure GCN and flat actor–critic policies:** From-scratch DDPG
 808 and TD3 quantify the benefit of anchoring; SAC and PPO provide
 809 secondary off-policy stochastic and on-policy comparisons. Residual
 810 SAC/PPO are included only after their residual-space probability cal-
 811 culations and matched flat variants pass the same validation protocol.

812 In addition, graph ablation experiments are planned to evaluate the role
 813 of different edge types. Specifically, we consider removing capacity-sharing
 814 edges and removing resource-sharing edges to assess whether these relational
 815 structures contribute to policy performance under disruption and demand
 816 uncertainty.

817 All methods receive the same decision-time observations and are eval-
 818 uated on identical paired random-number streams. In demand-prior-drift
 819 experiments, heuristics receive the same misspecified demand prior available
 820 to the controller and no method receives the realized future demand rates.
 821 Learned policies use the same training-scenario distribution, environment-

Table 5: Final comparison hierarchy for the proposed controller.

Comparison layer	Required policies	Question answered
Anchor improvement	im- AFR-GCN-DDPG versus MDL-2	Does the learned correction improve the strong policy from which it starts?
Strong heuristics	heuristic- ISO, MYO, F-MYO, uMYO, pMYO, MDL-1, and MDL-2	Is the controller competitive with sharing, forecast-aware, patient-aware, and look-ahead rules?
Representation ablation	AFR-GCN-DDPG versus matched AFR-flat-DDPG	Is any gain attributable to graph representation rather than residual learning alone?
Backbone robustness	ro- Residual GCN-DDPG versus residual GCN-TD3, each with a matched flat counterpart	Does the result persist when the deterministic actor-critic backbone changes?
Anchor ablation	abla- Residual versus pure GCN/flat DDPG and TD3	Does the heuristic anchor improve sample efficiency, stability, and final performance?
Deployment reference	MDL-2/pMYO rollout shields and supervised GCN shield selector	How does policy quality trade off against online simulation cost, and can RL outperform non-RL distillation?
Secondary family	GCN/flat SAC and PPO; residual forms after matched implementation	Is the conclusion robust to stochastic off-policy and on-policy learning?
Graph mechanism	Full graph, no/shuffled edges and geographic-edge ablations	Does the policy use topology rather than merely additional parameters?

822 interaction budget, checkpoint-selection budget, and held-out evaluation seeds
 823 within each matched comparison. Online decision time and the number of
 824 simulator rollouts are reported separately for rollout-based methods. This
 825 prevents a performance difference from being created by unequal forecasts,
 826 simulator access, evaluation noise, or hidden online computation.

827 5.2.2. *Performance Metrics*

828 We evaluate each method using both objective-based and engineering-
 829 relevant performance metrics. The primary metric is the average total weighted
 830 objective J_{tot} in Equation (8). It includes the complete operating-cost group
 831 (reagent purchase, holding, and shortage; bioreactor holding and shortage;
 832 and reagent and capacity transfer) and the patient-related penalty group
 833 (patient loss, material or therapy expiry, and urgency). The percentage im-
 834 provement reported in headline comparisons always uses the anchor’s com-
 835 plete J_{tot} as the denominator, as in Equation (9); we do not remove large
 836 components after observing the result.

837 Because a weighted total can be dominated by patient-loss or capacity-
 838 shortage penalties, every formal comparison also reports the absolute paired
 839 difference and an additive cost decomposition. Component-specific percent-
 840 ages are secondary mechanism measures: they reveal whether a policy saves
 841 purchase, holding, shortage, transfer, or patient-penalty cost, but do not
 842 replace the primary total-objective comparison. When an oracle or other
 843 defensible lower-cost reference is evaluated, we also report the headroom-
 844 capture measure in Equation (10). To strengthen the application relevance
 845 of the experiments, we track the following secondary metrics:

- 846 • **Cost decomposition:** operating/base cost and its reagent, bioreac-

847 tor, and transfer components, together with patient-loss, expiry, and
848 urgency penalties.

- 849 • **Completion service level:** the fraction of eligible patient demand
850 completed within the planning horizon.
- 851 • **Patient loss and manufacturing-period ineligibility:** the number
852 of patients who become ineligible or expire before treatment and the
853 fraction who become ineligible after manufacturing begins.
- 854 • **Average waiting time / fulfillment delay:** the average number of
855 epochs that patient specimens wait before manufacturing begins.
- 856 • **Reagent shortage frequency:** the frequency with which manufac-
857 turing is delayed because insufficient reagents are available.
- 858 • **Bioreactor utilization:** the fraction of available bioreactor capacity
859 used for production across facilities.
- 860 • **Transshipment and relocation activity:** the total number of reagent
861 movements and idle-bioreactor relocations across the network; speci-
862 men movements remain zero under the autologous identity constraint.
- 863 • **Inference time:** the computational time required for a trained policy
864 or heuristic to generate one epoch-level decision.

865 These metrics are included because the objective of PRM capacity plan-
866 ning is not only to reduce a modeled weighted objective, but also to support
867 timely patient-specific manufacturing, effective resource utilization, and reli-
868 able system performance under demand and supply uncertainty. Cost-weight

869 sensitivity is therefore required before translating an objective difference into
870 an economic claim.

871 5.2.3. *Heuristic Implementations*

872 We provide brief descriptions of the heuristic methods implemented:

- 873 • **MYO (Myopic):** MYO focuses on minimizing costs within the cur-
874 rent period without considering future consequences.
- 875 • **MDL-1 and MDL-2:** These methods extend the planning horizon to
876 one and two lookahead periods, respectively, using deterministic state
877 transitions based on mean demand.
- 878 • **ISO (Isolated Regional Facilities):** ISO assumes no sharing of re-
879 sources between manufacturing facilities and primarily concentrates on
880 reagent replenishment.

881 5.2.4. *Implementation and statistical protocol*

882 The models are implemented in Python using PyTorch, and all experi-
883 ment parameters, training seeds, checkpoint candidates, deployment scales,
884 and evaluation streams are stored in configuration files. Within a matched
885 comparison, the graph and flat policies receive the same state variables, an-
886 chor actions, teacher targets, environment interactions, validation budget,
887 and common-random-number (CRN) holdout episodes. The graph policy
888 replaces the flat encoder with message passing over the clinic graph; the re-
889 maining actor-critic and deployment components are held fixed or parameter
890 matched.

891 The replenishment-residual DDPG and TD3 confirmation uses 300 train-
892 ing episodes for each of five seeds, checkpoint candidates between episodes
893 50 and 300, and 100 independent paired holdout replications per seed. Re-
894 ported confidence intervals use a two-level bootstrap over training seeds and
895 paired evaluation replications. The regional network-action experiment uses
896 three independently initialized graph and flat policies, 30 paired validation
897 episodes, and 100 previously unseen paired holdout replications per seed.
898 Because its selected checkpoint is dominated by advantage-filtered teacher
899 distillation, that experiment is interpreted as graph-policy evidence rather
900 than as independent evidence for online DDPG updates.

901 The conservative online-TD3 screen starts from the corresponding frozen
902 pretrained graph and flat checkpoints and runs 100 online episodes for seeds
903 0, 1, and 2. Each run completed 5,200 update calls, including 5,073 critic
904 updates and 2,286 delayed actor updates. Final and frozen checkpoints are
905 evaluated with identical CRN keys, using 20 validation and 100 holdout
906 replications per seed for the final policies and 5 validation and the same 100
907 holdout replications per seed for the frozen controls. This campaign was run
908 on an NVIDIA RTX 4090 GPU; all optimization diagnostics were finite, and
909 no CPU fallback occurred. These experiment-specific budgets replace the
910 earlier generic one-million-step description and make the reported results
911 traceable to the executed configurations.

912 *5.3. distributed manufacturing simulation*

913 To evaluate the proposed method, we now construct a simulation en-
914 vironment of a decentralized manufacturing system consisting of $N = 20$
915 manufacturing facilities and its dedicated reagent supplier. Each PRM prod-

Algorithm 2 State normalization

Require: State variables: s_t^n , r_t^n , b_t^n , and p_t^n

1: Maximal values of the state variables for all facilities: M_s , M_r , M_b , and M_p

Ensure: Normalized state variables: $\hat{s}_t^n$, $\hat{r}_t^n$, $\hat{b}_t^n$, $\hat{p}_t^n$

2: **for** $n = 1$ to N **do**

3: $\hat{s}_t^n = s_t^n / M_s$

4: $\hat{r}_t^n = r_t^n / M_r$

5: $\hat{b}_t^n = b_t^n / M_b$

6: $\hat{p}_t^n = p_t^n / M_p$

7: **end for**

916 uct requires $T = 3$ epochs of manufacturing time, and the considered finite
917 problem horizon contains $\Gamma = 52$ decision epochs. For every epoch t , there
918 will be d_t^n new demand for facility n . The demand arrival process for facility
919 n follows a Poisson distribution $\text{Poisson}(\lambda_n)$. The supplier disruption follows
920 independent Bernoulli processes $\text{Ber}(\zeta_n)$, i.e., the facility n has a probability
921 ζ_n to have supplier disruption and could not purchase any reagent for the
922 epoch. This simulation can be further adjusted to reflect any real situation,
923 e.g., supply disruption and demand uncertainty, that may be encountered.
924 Such flexibility in using the simulation system will be demonstrated in Sec-
925 tion 5.4 when we compare GCN-DDPG with other methods.

926 Patient-condition parameters are calibrated before policy comparison.
927 The nominal scenario uses $\lambda_H = 0.0013$, $\lambda_F = 0.0156$, Weibull shape $\kappa = 1.5$
928 and scale $\theta = 4$, post-deterioration acceleration $\rho = 3$, eligibility threshold
929 $\epsilon = 0.80$, and waiting-time pressure $0.0013\alpha^2$. We selected these values from

930 a fixed-seed sweep whose pre-specified target was a 10–15% *patient ineligi-*
931 *bility during manufacturing* rate over the three-week process. This outcome
932 is not labeled a technical manufacturing failure: it records a patient who
933 becomes clinically ineligible after production begins, whereas a process or
934 release-specification failure would be a separate stochastic event. The target
935 is consistent with the order of magnitude of pre-infusion attrition reported for
936 autologous CAR-T pathways [3] and with a 5% weekly eligibility-transition
937 benchmark, which gives $1 - (1 - 0.05)^3 = 14.26\%$ over three weeks.

938 The patient-facing weights are $\omega_{\text{lost}} = 500,000$, $\omega_{\text{exp}} = 100,000$, and $\omega_{\text{urg}} =$
939 $25,000$. The earlier exploratory values (50,000, 40,000, 5,000) made one-time
940 patient loss cheaper than even one period of combined reagent and bioreactor
941 shortage, so a policy could lower its objective when a patient left the system.
942 The calibrated ordering removes this directional inconsistency and is held
943 fixed for all policy comparisons.

944 The epoch is assumed to occur weekly. Reagent and idle-bioreactor trans-
945 fers enter an in-transit pipeline and arrive after one, two, or three epochs,
946 according to whether the geographic distance is at most 500 miles, between
947 500 and 1,500 miles, or above 1,500 miles. Patient specimens remain at
948 their enrolling clinic, so $w_t^n = 0$ in the patient-condition experiments. Af-
949 ter demand and any due transfers have arrived, the next observed state is
950 $\mathbf{x}_{t+1}^n = (s_{t+1}^n, r_{t+1}^n, \mathbf{b}^n, P_{t+1})$ for facility $n = 1, \dots, N$, according to Equation
951 (1). The state variables are updated by the following rules:

$$\begin{aligned}
s_{t+1}^n &= s_t^n - m_t^n + d_t^n + w_t^n \\
r_{t+1}^n &= r_t^n - m_t^n + e_t^n + p_t^n \\
b_{t+1}^{n,\tau} &= \begin{cases} b_t^{n,0} - m_t^n + b_t^{n,1} + q_t^n & \text{if } \tau = 0 \\ b_t^{n,\tau+1} & \text{if } 1 \leq \tau \leq T-2 \\ m_t^n & \text{if } \tau = T-1 \end{cases}
\end{aligned}$$

952 where $m_t^n = \min(s_t^n, b_t^{n,0}, r_t^n)$ is the number of products that begin man-
 953 ufacturing at epoch t , d_t^n is the demand arriving at facility n in epoch t , and
 954 P_{t+1} follows an exogenous Markov chain from P_t . The operating component
 955 of the single-period objective can be decomposed as

$$\begin{aligned}
c_{\text{op}}(\mathbf{A}_t; \mathbf{X}_t) &= \sum_{n=1}^N C_{\text{op},t}^n, \\
C_{\text{op},t}^n &= C_{\text{Reg},t}^n + C_{\text{Bio},t}^n + C_{\text{Tran},t}^n.
\end{aligned}$$

956 The meanings of $C_{\text{Reg},t}^n$, $C_{\text{Bio},t}^n$, and $C_{\text{Tran},t}^n$ are summarized below. These
 957 terms constitute only c_{op} ; the patient-related terms in Equation (7) must
 958 also be included to obtain the total weighted objective.

- 959 • $C_{\text{Reg},t}^n = c_R p_t^n + h_R (r_{t+1}^n - s_{t+1}^n)^+ + p_R (s_{t+1}^n - r_{t+1}^n)^+$ is the cost related
 960 to reagent at epoch t , including the reagent replenishment cost $c_R p_t^n$,
 961 the reagent overstock holding cost $h_R (r_{t+1}^n - s_{t+1}^n)^+$, and the reagent
 962 understock penalty $p_R (s_{t+1}^n - r_{t+1}^n)^+$;
- 963 • $C_{\text{Bio},t}^n = h_B (b_{t+1}^{n,0} - s_{t+1}^n)^+ + p_B (s_{t+1}^n - b_{t+1}^{n,0})^+$ is the cost related to

964 bioreactor at epoch t , including the bioreactor overstock holding cost
965 $h_B (b_{t+1}^{n,0} - s_{t+1}^n)^+$, and the bioreactor understock penalty $p_B (s_{t+1}^n - b_{t+1}^{n,0})^+$;
966 • $C_{\text{Tran},t}^n = K_R |e_t^n| + K_B |q_t^n| + K_S |w_t^n|$ is the generic transfer-cost expres-
967 sion. In the patient-condition experiments $w_t^n = 0$, so the specimen
968 term vanishes and only reagent transfer and idle-bioreactor relocation
969 remain.

Table 6: Modeled cost coefficients used in the calibrated experiments. Values are weighted-objective units and are held fixed across policies. Transfer coefficients are base values before the geographic distance and travel-time surcharge. Patient-related coefficients are penalty or shadow-value parameters rather than observed cash expenditures.

Component	Coefficient	Interpretation
Reagent purchase, c_R	42,174.0	per purchased unit
Reagent holding, h_R	113.5	per excess unit-period
Reagent shortage, p_R	86,504.5	per shortage unit-period
Bioreactor holding, h_B	14.4	per idle unit-period
Bioreactor shortage, p_B	50,273.6	per shortage unit-period
Reagent transfer, K_R	200.0	per transferred unit, before surcharge
Capacity transfer, K_B	500.0	per relocated unit, before surcharge
Patient loss, ω_{lost}	500,000	per lost patient
Expiry, ω_{exp}	100,000	per wasted material or therapy unit
Urgency, ω_{urg}	25,000	per at-risk unserved patient-period

970 5.4. Case Study 1: Interconnected Clinic Network Analysis

971 This case study holds the demand prior equal to the arrival distribution
972 and varies the supplier disruption rate over 0.05, 0.30, and 0.60. Its purpose

973 is to measure the value of graph-aware coordination when the determinis-
 974 tic heuristics are not disadvantaged by demand misspecification. The final
 975 matched multi-seed campaign for these three regimes is still pending; conse-
 976 quently, we do not report numerical disruption-effect claims in the present
 977 draft. This case study will use the same five training seeds, paired Monte
 978 Carlo evaluation, checkpoint selection, and two-level bootstrap protocol as
 979 Case Study 2.

980 *5.5. Case Study 2: Adaptability to Non-Stationary Demand*

981 This case study evaluates the main robustness hypothesis under patient-
 982 condition and geographic network dynamics. The 20 clinics are partitioned
 983 into four five-clinic demand groups. MDL-2 plans with Poisson prior rates
 984 $[7.5, 3.2, 6.0, 3.5]$, whereas the simulator generates arrivals using true rates
 985 $[10.5, 3.52, 7.5, 3.15]$. The environment also includes patient deterioration,
 986 expiry, clustered demand shocks, regional supplier disruptions, and distance-
 987 dependent transfer lead times. During training, demand rates are randomized
 988 between 0.8 and 1.5 of the scenario rates and forecast error between 0 and
 989 0.35; evaluation is fixed to the scenario above.

990 Before retraining learned policies, we evaluated ISO and MDL-2 with 100
 991 paired Monte Carlo replications to verify that the calibrated patient pro-
 992 cess and objective produce the intended clinical range. Table 7 reports this
 993 calibration pilot and the subsequent matched learned-policy screen. Learned-
 994 policy results produced before the complete patient lifecycle and objective
 995 calibration are not comparable and are therefore retired rather than carried
 996 forward into this experiment.

997 Both heuristics fall inside the pre-specified clinical calibration band: manufacturing-

Table 7: Calibrated demand-prior-drift results. ISO and MDL-2 use the initial 100-replication calibration stream; learned-policy and matched heuristic rows aggregate 100 paired replications for each of three training seeds. Objective values are the total modeled weighted objective in billions of cost units, not realized accounting expenditure. Lower objective, patient loss, and manufacturing-period ineligibility are better; higher completion service and eligibility are better.

Method	Objective (10^9 units)	Completion	Eligibility	Lost	Mfg. inelig.
Initial ISO calibration	3.990456	0.337612	0.377537	4021.92	0.146398
Initial MDL-2 calibration	3.895694	0.371948	0.421024	3698.87	0.121364
AFR-Flat-DDPG	3.927584	0.370359	0.507030	3730.36	0.122223
AFR-GCN-DDPG	3.927594	0.370359	0.507030	3730.35	0.122223
AFR-GCN-TD3	3.927628	0.370358	0.507030	3730.36	0.122223
MDL-2	3.927620	0.370359	0.507030	3730.36	0.122223
ISO	4.011547	0.337521	0.460044	4043.10	0.146468
pMYO	4.180485	0.343311	0.496776	3894.09	0.133332
MYO	4.194549	0.342889	0.496586	3897.01	0.133582

period patient ineligibility is 14.64% for ISO and 12.14% for MDL-2. MDL-2 reduces the mean total weighted objective by 94.76 million modeled cost units (2.37%) relative to ISO; the paired 95% interval for MDL-2 minus ISO is $[-108.81, -80.71]$ million cost units, and MDL-2 has a lower objective in 92 of 100 paired replications. It also loses 323.05 fewer patients on average, discards 49.40 fewer in-process therapies, and raises completion service by 0.03434.

The calibrated learned-policy screen used 100 training episodes and three training seeds. AFR-GCN-DDPG selected and deployed a nonzero residual in all three seeds; AFR-GCN-TD3 and AFR-Flat-DDPG each returned to MDL-2 in one seed. AFR-GCN-DDPG reduced the mean total weighted objective relative to MDL-2 by only 26,519 modeled cost units (0.000675%),

1010 with a two-level bootstrap 95% interval of $[-89,588, 40,330]$ cost units. Its
1011 mean objective was 9,460 cost units higher than the matched flat residual pol-
1012 icy, with interval $[-7,951, 24,621]$. Completion service and patient loss were
1013 effectively unchanged. Thus the screen demonstrates safeguard-compatible
1014 parity with a strong anchor, but it does not support either a material RL
1015 improvement or a graph-representation advantage. The accepted correction
1016 primarily substitutes additional reagent purchase and holding cost for lower
1017 reagent shortage cost. Because this residual changes replenishment only, the
1018 next matched experiment expands the bounded correction to distance-aware
1019 reagent and idle-bioreactor transfers, where geographic message passing can
1020 affect the decision.

1021 The small headline percentage in this screen is intentionally computed
1022 against the complete weighted objective rather than a narrower, post hoc
1023 denominator. Patient-related penalties and capacity-shortage penalties can
1024 dominate that total, while a replenishment-only residual directly changes
1025 only a subset of the operating components. We therefore interpret 0.000675%
1026 as statistical parity rather than as a material economic saving. Subsequent
1027 network-residual comparisons report the total-objective change together with
1028 operating, bioreactor-shortage, transfer, and patient-penalty decompositions
1029 so that denominator dilution and the mechanism of any improvement remain
1030 visible.

1031 5.5.1. Formal five-seed demand-prior-drift confirmation

1032 The formal replenishment-residual comparison increased the training bud-
1033 get to 300 episodes and used five independent training seeds. Each selected
1034 checkpoint was evaluated on 100 paired holdout replications, yielding 500

paired outcomes per method. Table 8 reports the mean outcomes, and Table 9 reports paired differences. The learned-policy values in these tables supersede the 100-episode diagnostic screen in Table 7.

Table 8: Five-seed demand-prior-drift confirmation. Objective values are the total modeled weighted objective in billions of units. Each row aggregates 100 paired holdout replications for each of five training seeds.

Method	Objective (10^9 units)	Completion	Eligibility	Waiting	Lost	At-risk unserved
MDL-2	1.268257	0.429982	0.530313	3.052625	3665.59	5206.41
AFR-Flat-DDPG	1.268089	0.430320	0.530523	3.052012	3663.28	5203.86
AFR-GCN-DDPG	1.268049	0.430321	0.530524	3.052012	3663.27	5203.86
AFR-GCN-TD3	1.268241	0.430317	0.530523	3.052019	3663.30	5203.88

Table 9: Paired five-seed demand-prior-drift comparisons. Differences are candidate minus baseline, so a negative objective difference favors the first method. Confidence intervals use the two-level bootstrap over training seeds and paired replications.

Comparison	ΔJ_{tot} (units)	Relative gap	Paired 95% CI	Wins / 500
AFR-GCN-DDPG vs. MDL-2	-207,485	-0.016360%	$[-332,641, -91,367]$	323
AFR-Flat-DDPG vs. MDL-2	-167,610	-0.013216%	$[-286,899, -46,230]$	319
AFR-GCN-DDPG vs. AFR-Flat-DDPG	-39,876	-0.003145%	$[-89,702, 14,025]$	285
AFR-GCN-TD3 vs. MDL-2	-15,823	-0.001248%	$[-142,607, 99,437]$	268
AFR-GCN-DDPG vs. AFR-GCN-TD3	-191,662	-0.015112%	$[-234,659, -148,401]$	377

AFR-GCN-DDPG improves on MDL-2 in every training-seed mean and has a two-level confidence interval fully below zero. Completion service increases by 0.000340, eligibility by 0.000211, average waiting time falls by 0.000612, patients lost fall by 2.32, and at-risk unserved patients fall by 2.55. This supports a statistically stable anchor improvement under demand-prior drift, but not a large economic effect. The matched flat policy also improves on MDL-2, and the direct graph-versus-flat interval includes zero. Replen-

1045 ishment residual learning is therefore supported in this scenario, whereas an
1046 independent graph-representation advantage is not.

1047 The matched TD3 policy preserves the patient-facing improvements but
1048 has a cost interval crossing zero. DDPG is lower cost than TD3 in all five
1049 training-seed means, and their paired interval is fully below zero. This
1050 comparison supports DDPG as the canonical backbone for the proposed
1051 replenishment-residual controller; it is not a claim that DDPG generally
1052 dominates TD3 outside this residual-learning design.

1053 *5.6. Case Study 3: Graph Attribution Under Regional Non-Stationarity*

1054 The third case study combines patient-condition dynamics, clinic geog-
1055 raphy, distance-based transfer delays, regional demand heterogeneity, and
1056 demand regime changes. Unlike the replenishment-only experiment, the net-
1057 work residual contains a node head for signed replenishment corrections and
1058 edge heads for reagent and idle-bioreactor transfer corrections. Edge flows
1059 are projected to preserve resource conservation and feasibility, while specimen
1060 transfer is masked to zero. This action space gives graph message passing a
1061 direct route to affect geographically structured decisions.

1062 *5.6.1. Network AFR-GCN-DDPG under an abrupt regional shift*

1063 The abrupt-shift policy uses transfer-focused advantage-filtered targets
1064 and a fixed deployment scale selected on 30 paired validation episodes. Three
1065 independently initialized graph policies and parameter-matched flat policies
1066 were then evaluated on the same 100 previously unseen paired holdout repli-
1067 cations per seed. The graph and flat models contain 524,775 and 528,404
1068 trainable parameters, respectively, so the comparison does not give the graph

1069 policy a parameter-count advantage.

Table 10: Three-seed abrupt-regional-shift comparisons. Objective differences are in millions of modeled weighted-objective units. Each comparison contains 300 paired holdout outcomes.

Comparison	ΔJ_{tot} (10^6 units)	Relative gap	Paired 95% CI	Δ completion	Δ patients lost
Network AFR-GCN-DDPG vs. MDL-2	-25.39	-1.1337%	[-29.76, -20.95]	+0.001357	-0.70
Network AFR-GCN-DDPG vs. AFR-Flat-DDPG	-25.01	-1.1166%	[-29.35, -20.61]	+0.002836	-9.86

1070 All three seed-level objective means favor the graph controller. Relative
 1071 to MDL-2, the completion-service 95% interval is [0.000240, 0.002458] and
 1072 the patients-lost interval is [-6.42, 5.07]; aggregate patient-loss noninferior-
 1073 ity therefore passes, but the strict per-seed clinical gate passes for two of
 1074 three seeds. Relative to the matched flat policy, the completion interval is
 1075 [0.001766, 0.003898] and the patients-lost interval is [-15.24, -4.31]. This
 1076 is the current statistically significant evidence for a graph-specific benefit:
 1077 it appears when the action space includes geographically structured edge
 1078 decisions, not when the actor modifies replenishment alone.

1079 Table 11 explains why the total-objective percentage is smaller than the
 1080 change in the component directly affected by the policy. MDL-2’s mean
 1081 objective is 2.2399 billion units. Bioreactor shortage, patient loss, and expiry
 1082 account for 44.10%, 43.53%, and 8.71%, respectively. The graph controller
 1083 reduces the bioreactor-shortage component by 25.42 million units, or 2.57%
 1084 of that component, which becomes a 1.13% improvement when divided by
 1085 the complete objective. The component percentage is a mechanism measure;
 1086 the complete-objective percentage remains the headline result.

1087 This positive network checkpoint is driven primarily by advantage-filtered
 1088 teacher distillation. Earlier unconstrained offline DDPG and TD3 updates

Table 11: Cost decomposition for MDL-2 and Network AFR-GCN-DDPG in the abrupt-regional-shift holdout. The final column is the within-component percentage change. Components are grouped without changing the total-objective denominator used in Table 10.

Component	MDL-2 mean (10^6 units)	Share of total	AFR change (10^6 units)	Component change
Bioreactor shortage	987.891	44.10%	−25.420	−2.57%
Patient loss	975.120	43.53%	−0.352	−0.04%
Expiry	195.008	8.71%	−0.084	−0.04%
Urgency	6.118	0.27%	+0.641	+10.48%
Other operating components	75.742	3.38%	−0.178	−0.23%
Total weighted objective	2239.879	100.00%	−25.393	−1.13%

weakened the policy or violated patient safeguards. The experiment therefore establishes value from the graph-aware AFR controller, but it does not by itself establish that online DDPG updates generated that value.

5.6.2. Conservative online-TD3 attribution screen

To test whether actual actor-critic updates add value beyond the frozen pretrained controller, we initialized matched graph and flat residual TD3 policies from their corresponding pretrained checkpoints and ran 100 online episodes for each of three seeds. Every run completed nonzero critic and actor updates with finite diagnostics. Final and frozen checkpoints were evaluated on identical 100-replication CRN holdouts.

Table 12: Final conservative TD3 outcomes under geographic regional demand drift. Values aggregate three training seeds and 100 paired holdout replications per seed.

Method	Objective (10^9 units)	Completion	Patients lost	Mfg. ineligibility
MDL-2	2.458807	0.490786	2211.86	0.073077
Final AFR-Flat-TD3	2.460346	0.490433	2213.75	0.073209
Final AFR-GCN-TD3	2.456263	0.491636	2206.70	0.072685

The final graph policy improves on MDL-2 for each seed (−0.051936%,

Table 13: Paired final-checkpoint TD3 comparisons. Objective differences are in millions of modeled weighted-objective units.

Comparison	ΔJ_{tot}	Relative gap	Paired 95% CI	Δ completion	Δ lost	Δ mfg. inelig.
Final AFR-GCN-TD3 vs. MDL-2	-2.544	-0.103459%	$[-3.872, -1.131]$	+0.000850	-5.16	-0.000392
Final AFR-Flat-TD3 vs. MDL-2	+1.539	+0.062595%	$[+0.807, +2.387]$	-0.000353	+1.89	+0.000132
Final AFR-GCN-TD3 vs. AFR-Flat-TD3	-4.083	-0.165951%	$[-5.633, -2.436]$	+0.001203	-7.05	-0.000524

1100 -0.139249% , and -0.119193%), deploys nonzero residuals in 73.50%, 47.63%,
 1101 and 67.25% of evaluated decisions, and passes all pre-specified clinical non-
 1102 inferiority checks. The matched flat policies are more expensive than MDL-2
 1103 and fail clinical noninferiority for all three seeds. The final graph-versus-flat
 1104 comparison is therefore statistically and clinically favorable.

1105 The online-learning attribution is more limited. The pooled final-minus-
 1106 frozen graph difference is -0.350 million units, with 95% interval $[-1.032, 0.155]$
 1107 million, and the GCN-minus-flat difference-in-differences is -0.485 million
 1108 units, with interval $[-1.489, 0.173]$ million. Both point estimates favor online
 1109 TD3 updates, but both intervals include zero. Thus the final graph controller
 1110 outperforms MDL-2 and matched flat TD3 in this regional screen, while the
 1111 independent incremental benefit of online TD3 over frozen pretraining is not
 1112 yet established. A five-scenario zero-shot evaluation and additional training
 1113 seeds are reserved for confirmation rather than being reported as completed
 1114 evidence here.

1115 6. Discussion and Conclusions

1116 This study develops a patient-condition and geography-aware simulation
 1117 of a 20-clinic PRM network and a residual GCN-DDPG controller that learns
 1118 around a strong MDL-2 anchor. Advantage-filtered distillation identifies

1119 sparse state-dependent corrections, while checkpoint selection, residual-scale
1120 calibration, and fallback prevent an unsupported correction from being de-
1121 ployed. This structure preserves the interpretability of the anchor while
1122 allowing the graph actor to respond to patient pressure, transfer pipelines,
1123 regional disruptions, and demand-prior drift.

1124 The calibrated patient process places both ISO and MDL-2 inside a clini-
1125 cally interpretable manufacturing-period ineligibility band, and the weighted
1126 objective no longer rewards early patient loss. Under demand-prior drift, the
1127 formal five-seed result shows that AFR-GCN-DDPG improves on MDL-2
1128 with a two-level confidence interval fully below zero while weakly improv-
1129 ing all reported patient-facing metrics. The effect is small, and a matched
1130 flat residual policy also improves on MDL-2; consequently, this experiment
1131 supports anchored residual learning but not an independent graph advan-
1132 tage. The matched backbone comparison favors DDPG over TD3 for this
1133 replenishment-residual problem, which supports DDPG as the canonical pro-
1134 posed instantiation without implying that it is universally superior.

1135 The graph-specific result emerges after the residual action is aligned with
1136 the network representation. Under an abrupt regional demand shift, Network
1137 AFR-GCN-DDPG significantly improves on both MDL-2 and a parameter-
1138 matched flat policy while improving completion service. The cost decompo-
1139 sition shows that the controller primarily reduces bioreactor-shortage penal-
1140 ties; reporting this component change alongside the full-objective percent-
1141 age explains the mechanism without changing the headline denominator. A
1142 separate conservative online-TD3 screen likewise yields a final graph pol-
1143 icy that outperforms MDL-2 and matched flat TD3 while satisfying clinical

1144 noninferiority. However, the pooled final-versus-frozen and graph-minus-flat
1145 difference-in-differences intervals both cross zero. We therefore distinguish
1146 the supported claim that the deployed graph-aware AFR controller creates
1147 value from the unresolved claim that online actor-critic updates indepen-
1148 dently create that value.

1149 Several limitations remain. The formal demand-prior-drift DDPG experi-
1150 ment has five training seeds, but the network-action DDPG and conservative
1151 TD3 studies currently have three. The abrupt-shift graph checkpoint is dom-
1152 inated by advantage-filtered teacher distillation, and cross-scenario zero-shot
1153 robustness is not yet established. The full disruption-rate, robust/rolling
1154 MDL-2, pure-policy, SAC, and PPO comparisons still require a common
1155 locked budget. The clinical calibration target is literature aligned but is not
1156 fitted to patient-level data from a single indication, so sensitivity bands re-
1157 main necessary. The patient-loss, expiry, urgency, and shortage coefficients
1158 are modeled decision weights rather than a complete accounting valuation;
1159 cost-weight sensitivity and external economic calibration are required before
1160 interpreting objective differences as realized monetary savings. External clin-
1161 ical data will also be needed to validate patient-risk, demand, disruption, and
1162 transfer-time assumptions before operational use.

1163 **Future Work** This study motivates several extensions. Patient condi-
1164 tion can be represented with longitudinal clinical covariates rather than the
1165 current eligibility and survival-risk summary. Raw-material quality, product
1166 quality, process failures, machine breakdowns, and operator availability can
1167 be added to the simulator. The immediate empirical priorities are to finish
1168 the locked five-scenario zero-shot evaluation, add training seeds 3 and 4 to the

1169 surviving network controller, and evaluate robust and rolling-prior MDL-2
1170 controls under the same CRN protocol. If online-learning attribution remains
1171 inconclusive, future training should use multi-scenario replay and a conserva-
1172 tive, explicitly anchor-relative actor objective rather than merely increasing
1173 the episode budget. Longer-term experiments should test unseen clinic net-
1174 works and transfer disruptions and should integrate production scheduling,
1175 product preparation and distribution, and treatment operations.

1176 **Data Availability**

1177 The data used in this study are generated from a simulation model.
1178 The simulation parameters and experimental settings are described in the
1179 manuscript. Code and additional materials are available from the corre-
1180 sponding author upon reasonable request.

1181 **Declaration of Competing Interest**

1182 The author declares that there is no known competing financial interest or
1183 personal relationship that could have appeared to influence the work reported
1184 in this paper.

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